# III.2 Standard and Canonical Forms of an LP problem

**Standard Form:** An LP-problem is said to be in **standard (inequality) form**, if its matrix representation is of the form

$$\boxed{\text{Maximize}} \qquad f(\mathbf{x}) = \mathbf{c}^T\mathbf{x}$$

$$\text{Subject to} \qquad A\mathbf{x} \le \mathbf{b}$$

"$\le$" inequality

$$\mathbf{x} \ge \mathbf{0}$$

all variables are $\geq 0$

*e.g.* Examples LP1, LP2, LP3 and LP5 above are in standard form.

Examples LP4 and LP6 are NOT in standard form.

Lets look at some examples to bring a problem to standard form.

**Example S1:** Change the LP problem to standard (inequality) form:

Minimize $f(x_1, x_2, x_3) = 2x_1 - 3x_2 - 6x_3$ (1)

subject to $7x_1 + 5x_2 - 6x_3 \geq -10$ (2)

$2x_1 + 8x_2 - 4x_3 \leq 3$ (3)

$9x_1 + 2x_2 + 5x_3 \geq -4$ (4)

$x_1 \leq 0$ (5)

$x_2, x_3 \geq 0$ (6)

There are 4 items to be fixed:

1. This is NOT a maximization problem
2. Inequalities (2) and (4) are NOT of type "$\leq$"
3. Inequality (5): The variable $x_1$ is NOT "$\geq$" [non-negative]

(1) Multiply the cost function by −1 to maximize

**Step 1:** Replace $f$ by $-f$ in (1) to obtain a maximization problem

| | | |
|---|---|---|
| Maximize | $-f(x_1, x_2, x_3) = -2x_1 + 3x_2 + 6x_3$ | (1) |
| subject to | $7x_1 + 5x_2 - 6x_3 \geq -10$ | (2) |
| | $2x_1 + 8x_2 - 4x_3 \leq 3$ | (3) |
| | $9x_1 + 2x_2 + 5x_3 \geq -4$ | (4) |
| | $x_1 \leq 0$ | (5) |
| | $x_2, x_3 \geq 0$ | (6) |

(2) multiply by −1.

**Step 2:** Multiply (2) and (4) by $-1$ to change $\geq$ to $\leq$:

| | | |
|---|---|---|
| Maximize | $-f(x_1, x_2, x_3) = -2x_1 + 3x_2 + 6x_3$ | (1) |
| subject to | $-7x_1 - 5x_2 + 6x_3 \leq 10$ | (2) |
| | $2x_1 + 8x_2 - 4x_3 \leq 3$ | (3) |
| | $-9x_1 - 2x_2 - 5x_3 \leq 4$ | (4) |
| | $x_1 \leq 0$ | (5) |
| | $x_2, x_3 \geq 0$ | (6) |

(3) change every $x_1$ to $-x_1$ and get

$-x_1 \leq 0 \Rightarrow x_1 \geq 0$

**Step 3** Fix (5): Change ALL $x_1$ to $-x_1$. We get

Maximize $g(x_1, x_2, x_3) = 2x_1 + 3x_2 + 6x_3$ (1)

subject to $7x_1 - 5x_2 + 6x_3 \le 10$ (2)

$-2x_1 + 8x_2 - 4x_3 \le 3$ (3)

$9x_1 - 2x_2 - 5x_3 \le 4$ (4)

$-x_1 \le 0$ (5)

$\Rightarrow x_1, x_2, x_3 \ge 0$ (6)

In matrix notation, If

$\vec{c}^T = [2 \ 3 \ 6]$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $A = \begin{bmatrix} 7 & -5 & 6 \\ -2 & 8 & -4 \\ 9 & -2 & -5 \end{bmatrix}$ $\vec{b} = \begin{bmatrix} 10 \\ 3 \\ 4 \end{bmatrix}$

then

$$\begin{cases} \text{Maximize} & \vec{c}^T \vec{x} \\ \text{subject to} & A\vec{x} \le \vec{b} \\ & \vec{x} \ge 0 \end{cases}$$

**Example S2:** Change the LP problem to standard (inequality) form:

| | | |
|---|---|---|
| Minimize | $3x_1 - x_2 + 2x_3$ | (1) |
| subject to | $2x_1 + 7x_2 - 6x_3 \le 5$ | (2) |
| | $x_1 - 8x_2 = 4$ | (3) |
| | $3x_1 - 2x_2 + 5x_3 \ge -12$ | (4) |
| | $x_1, x_2 \ge 0$ | (5) |

There are 4 items to be fixed:

1. This is NOT a maximization problem
2. (3) is an equality !!
3. Inequality (4) is NOT of type "$\le$"
4. $x_3$ is a *free* variable: (not $x_3 \ge 0$)

   ↑ not Constrained

(1) Multiply (1) by $-1$.

(2) change "$=$" to "$\le$" and "$\ge$"



(3) Multiply (4) by $-1$.

**Step 1:** Problems 1., 2., 3 above can be fixed independently:

$$\boxed{\text{Maximize}} \quad -3x_1 + x_2 - 2x_3 \qquad (1)$$

$$\text{subject to} \quad 2x_1 + 7x_2 - 6x_3 \le 5 \qquad (2)$$

$$\begin{cases} x_1 - 8x_2 \ge 4 & \xRightarrow[\text{multiply by } -1]{} \; -x_1 + 8x_2 \le -4 \qquad (3A) \\ x_1 - 8x_2 \le 4 & \qquad (3B) \end{cases}$$

$$-3x_1 + 2x_2 - 5x_3 \le 12 \qquad (4)$$

$$x_1,\, x_2 \ge 0 \qquad (5)$$

**Step 2:** Remove the free variable $x_3$: set $x_3 = z_1 - z_2$ $\quad z_1, z_2 \ge 0$

$$\text{Maximize} \quad -3x_1 + x_2 - 2(z_1 - z_2) \qquad (1)$$

$$\text{subject to} \quad 2x_1 + 7x_2 - 6(z_1 - z_2) \le 5 \qquad (2)$$

$$-x_1 + 8x_2 \le -4 \qquad (3A)$$

$$x_1 - 8x_2 \le 4 \qquad (3B)$$

$$-3x_1 + 2x_2 - 5(z_1 - z_2) \le 12 \qquad (4)$$

$$x_1,\, x_2,\, \underline{z_1, z_2} \ge 0 \qquad (5)$$

We get:

| | | |
|---|---|---|
| Maximize | $-3x_1 + x_2 - 2z_1 + 2z_2$ | (1) |
| subject to | $2x_1 + 7x_2 - 6z_1 + 6z_2 \le 5$ | (2) |
| | $-x_1 + 8x_2 \le -4$ | (3A) |
| | $x_1 - 8x_2 \le 4$ | (3B) |
| | $-3x_1 + 2x_2 - 5z_1 + 5z_2 \le 12$ | (4) |
| | $x_1, x_2, z_1, z_2 \ge 0$ | (5) |

In matrix form:

If $\vec{c}^T = [-3 \;\; 1 \;\; -2 \;\; 2]$ $\quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ z_1 \\ z_2 \end{bmatrix}$ $\quad A = \begin{bmatrix} 2 & 7 & -6 & 6 \\ -1 & 8 & 0 & 0 \\ 1 & -8 & 0 & 0 \\ -3 & 2 & -5 & 5 \end{bmatrix}$ $\quad \vec{b} = \begin{bmatrix} 5 \\ -4 \\ 4 \\ 12 \end{bmatrix}$

$$\begin{cases} \text{Maximize } \vec{c}^T \vec{x} \\ \text{Subject to } A\vec{x} \le \vec{b}, \quad \vec{x} \ge 0 \end{cases}$$

**Canonical Form:** An LP-problem is said to be in **canonical (equality) form**, if its matrix representation is of the form

$$\boxed{\text{Maximize}} \qquad f(\mathbf{x}) = \mathbf{c}^T\mathbf{x}$$

$$\text{Subject to} \qquad A\mathbf{x} = \mathbf{b}$$

$$\mathbf{x} \geq \mathbf{0}$$

If $\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ we require that $b_i \geq 0$ for all $i$

Where the entries $b_1, \ldots, b_n$ of the vector $\mathbf{b}$ are all non-negative.

**Remark:** In some textbooks this is called the standard form ! So it is better to use the phrases *'inequality form'* (instead of 'standard form') and *'equality form'* (instead of canonical form). Also, some textbooks require a minimization problem, not a maximization problem.

We can change an LP problem to canonical form by introducing so called **slack variables**:

**Example C1:** Change the LP problem to canonical (equality) form:

Maximize $f(x_1, x_2) = 2x_1 + 3x_2$ (1)

subject to $x_1 + x_2 = 10$ (2)

$7x_1 - 4x_2 \le 6$ (3)

$-2x_1 + 3x_2 \le -5$ (4)

$x_2 \ge 0$ (5)

① Make this non-negative Multiply by -1.

**Step 1:** Make all right-hand numbers non-negative: We multiply (4) by :

$2x_1 - 3x_2 \ge 5$ (4a)

**Step 2:** Make all inequalities '≤' to '=' :

② insert a slack variable

Consider (3): $7x_1 - 4x_2 \le 6$. We set $s = 6 - (7x_1 - 4x_2) \ge 0$. Then

$7x_1 - 4x_2 + s = 6, \qquad s \ge 0.$ (3a)

The new variable $s$ is called a ***slack variable***.

**Step 3:** Make all inequalities '≥' to '=' :

(3) insert a surplus variable.

Consider (4a): $2x_1 - 3x_2 \geq 5$. We set $e = (2x_1 - 3x_2) - 5 \geq 0$, and can write

$$2x_1 - 3x_2 - e = 5, \qquad e \geq 0. \qquad \text{(4b)}$$

The new variable $e$ is called a ***surplus variable***.

**Step 4:** In (5), $x_1$ is still a *free variable*. As before, we write

$$x_1 = z_1 - z_2, \qquad z_1, z_2 \geq 0 \qquad \text{(5a)}$$

Combining all the above, we obtain

| | | |
|---|---|---|
| Maximize | $f(z_1, z_2, x_2) = 2(z_1 - z_2) + 3x_2$ | (1) |
| subject to | $(z_1 - z_2) + x_2 = 10$ | (2) |
| | $7(z_1 - z_2) - 4x_2 + s = 6$ | (3a) |
| | $2(z_1 - z_2) - 3x_2 - e = 5$ | (4b) |
| | $z_1, z_2, x_2, s, e \geq 0$ | (5)+5(a) |

That is,

| | | |
|---|---|---|
| Maximize | $2z_1 - 2z_2 + 3x_2$ | (1) |
| subject to | $z_1 - z_2 + x_2 = 10$ | (2) |
| | $7z_1 - 7z_2 - 4x_2 + s = 6$ | (3a) |
| | $2z_1 - 2z_2 - 3x_2 - e = 5$ | (4b) |
| | $z_1, z_2, x_2, s, e \geq 0$ | (5)+5(a) |

In matrix form, if

$$\vec{c}^T = \begin{bmatrix} 2 & -2 & 3 & 0 & 0 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} z_1 \\ z_2 \\ x_2 \\ s \\ e \end{bmatrix} \quad A = \begin{bmatrix} 2 & -2 & 3 & 0 & 0 \\ 7 & -7 & -4 & 1 & 0 \\ 2 & -2 & -3 & 0 & -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 10 \\ 6 \\ 5 \end{bmatrix}$$

then

$$\begin{cases} \text{Maximize} & \vec{c}^T \vec{x} \\ \text{subject to} & A\vec{x} = \vec{b} \\ & \vec{x} \geq 0 \end{cases}$$

**Example C2:** Change the LP problem to standard (inequality) form:

Change to: Maximize

$$\boxed{\text{Minimize}} \quad 3x_1 - x_2 + 2x_3 \qquad (1)$$

insert slack variable

$$\text{subject to} \quad 2x_1 + 7x_2 - 6x_3 \le 5 \qquad (2)$$

$$-x_1 + 8x_2 = -4 \qquad (3)$$

← change to +4

change to +12

$$3x_1 - 2x_2 + 5x_3 \le -12 \qquad (4)$$

$$x_1,\ x_2 \ge 0 \qquad (5)$$

← $x_3$ is free

**Step 1:** Change to a maximization problem: Multiply (1) by :

$$\boxed{\text{Maximize}} \quad -3x_1 + x_2 - 2x_3 \qquad (1a)$$

**Step 2:** Make right-hand numbers non-negative: We multiply (3),(4) by :

$$x_1 - 8x_2 = 4 \qquad (3a)$$

$$-3x_1 + 2x_2 - 5x_3 \ge 12 \qquad (4a)$$

insert Surplus variable

**Step 3:** Make '$\le$' to '$=$' using slack variables: (2) becomes

$$2x_1 + 7x_2 - 6x_3 + s = 5, \qquad s \ge 0 \qquad (2a)$$

**Step 4:** Make '$\geq$' to '$=$' using surplus variables: (4a) becomes

$$-3x_1 + 2x_2 - 5x_3 - e = 12, \qquad e \geq 0 \qquad \text{(4b)}$$

**Step 5:** Fix all *free variables*. $x_3$ is a free variable, so we write

$$x_3 = z_1 - z_2, \qquad z_1, z_2 \geq 0 \qquad \text{(5a)}$$

Combining all the above, we obtain:

| | | |
|---|---|---|
| Maximize | $-3x_1 + x_2 - 2(z_1 - z_2)$ | (1a) |
| subject to | $2x_1 + 7x_2 - 6(z_1 - z_2) + s = 5$ | (2a) |
| | $x_1 - 8x_2 = 4$ | (3a) |
| | $-3x_1 + 2x_2 - 5(z_1 - z_2) - e = 12$ | (4b) |
| | $x_1,\ x_2,\ z_1,\ z_2,\ s,\ e \geq 0$ | (5b) |

That is,

Maximize $\quad -3x_1 + x_2 - 2z_1 + 2z_2 \qquad$ (1a)

subject to $\quad 2x_1 + 7x_2 - 6z_1 + 6z_2 + s = 5 \qquad$ (2a)

$$x_1 - 8x_2 = 4 \qquad (3a)$$

$$-3x_1 + 2x_2 - 5z_1 + 5z_2 - e = 12 \qquad (4b)$$

$$x_1, x_2, z_1, z_2, s, e \geq 0 \qquad (5b)$$

(One usually relabels the new variables:

$$z_1 \to x_3, \quad z_2 \to x_4, \quad s \to x_5, \quad e \to x_6)$$

In matrix form: LF

$$\vec{c}^T = [-3 \quad 1 \quad -2 \quad 2 \quad 0 \quad 0], \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ z_2 \\ z_3 \\ s \\ e \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 7 & -6 & 6 & 1 & 0 \\ 1 & -8 & 0 & 0 & 0 & 0 \\ -3 & 2 & -5 & 5 & 0 & -1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 5 \\ 4 \\ 12 \end{bmatrix}$$

than

$$\begin{cases} \text{Maximice } f(x_1, x_2) = \vec{c}^T x \\ \text{Subject to: } A\vec{x} = \vec{b} \\ \qquad\qquad\qquad \vec{x} \geq 0 \end{cases}$$

**Exercise S3:** Change the LP problem to

a) standard (inequality) form
b) canonical (equality) form

| | | |
|---|---|---|
| Minimize | $2x_1 - 3x_2 - 6x_3$ | (1) |
| subject to | $-7x_1 - 5x_2 + 6x_3 \geq -9$ | (2) |
| | $x_1 + x_3 = 3$ | (3) |
| | $9x_1 - 2x_2 - 5x_3 \leq -36$ | (4) |
| | $x_1 \leq 0, \quad x_2 \geq 0 \qquad x_3 \geq 0$ | (5) |

a) Standard form

1. Multiply (1) by -1
2. Multiply (2) by (-1)

Maximize $-2x_1 + 3x_2 + 6x_3$ (1)

Subject to

$$\begin{cases} 7x_1 + 5x_2 - 6x_3 \le 9 & (2a) \\ x_1 + x_3 = 3 & (3) \\ 9x_1 - 2x_2 - 5x_3 \le -36 & (4) \\ x_1 \ge 0, \quad x_2 \ge 0, \quad x_3 \ge 0 & (5) \end{cases}$$

slack variable

(3) change all $x_1$ to $-x_1$, split (3)

Maximize $2x_1 + 3x_2 + 6x_3$

Subject to:

$$\begin{cases} -7x_1 + 5x_2 - 6x_3 \le 9 \\ -x_1 + x_3 \le 3 \\ -x_1 + x_3 \ge 3 \implies x_1 - x_3 \le -3 \\ -9x_1 - 2x_2 - 5x_3 \le -36 \\ x_1 \ge 0, \quad x_2 \ge 0, \quad x_3 \ge 0 \end{cases}$$

$$\begin{cases} \text{Maximize} & 2x_1 + 3x_2 + 6x_3 \\ \text{Subject to:} & -7x_1 + 5x_2 - 6x_3 \le 9 \\ & -x_1 + x_3 \le 3 \\ & x_1 - x_3 \le -3 \\ & -9x_1 - 2x_2 - 5x_3 \le -36 \\ & x_1, x_2, x_3 \ge 0 \end{cases}$$

If $\vec{c}^T = \begin{bmatrix} 2 & 3 & 6 \end{bmatrix}$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $A = \begin{bmatrix} -7 & 5 & -6 \\ -1 & 0 & 1 \\ 1 & 0 & -1 \\ -9 & -2 & -5 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 9 \\ 3 \\ -3 \\ -36 \end{bmatrix}$

then
$$\begin{cases} \text{Maximize} & \vec{c}^T \vec{x} \\ \text{Subject to} & A\vec{x} \le \vec{b} \\ & \vec{x} \ge 0 \end{cases}$$

b) <u>Canonical form:</u>

$$\begin{cases} \text{Minimize} & 2x_1 - 3x_2 - 6x_3 & (1) \\ \text{Subject to:} & -7x_1 - 5x_2 + 6x_3 \geq -9 & (2) \\ & x_1 + x_3 = 3 & (3) \\ & 9x_1 - 2x_2 - 5x_3 \leq -36 & (4) \\ & x_1 \leq 0, \quad x_2, x_3 \geq 0 & (5) \end{cases}$$

① Multiply (1) by (-1) to Maximize,
Multiply (2) and (4) by -1 so $b_i \geq 0$:

$$\begin{cases} \text{Maximize} & -2x_1 + 3x_2 + 6x_3 \\ \text{Subject to:} & 7x_1 + 5x_2 - 6x_3 \leq 9 \\ & x_1 + x_3 = 3 \end{cases}$$

$$-9x_1 + 2x_2 + 5x_3 \geq 36$$

$$x_1 \leq 0, \quad x_2, x_3 \geq 0$$

② Insert slack / surplus variables

$$\begin{cases} \text{Maximize} & -2x_1 + 3x_2 + 6x_3 \\ \text{Subject to:} & 7x_1 + 5x_2 - 6x_3 + s = 9 \\ & x_1 + x_3 = 3 \\ & -9x_1 + 2x_2 + 5x_3 - e = 36 \\ & x_1 \leq 0, \; x_2, x_3, s, e \geq 0 \end{cases}$$

③ Replace $x_1$ by $-x_1$:

$$\begin{cases} \text{Maximize} & 2x_1 + 3x_2 + 6x_3 \\ \text{Subject to} & -7x_1 + 5x_2 - 6x_3 + s = 9 \\ & -x_1 + x_3 = 3 \\ & 9x_1 + 2x_2 + 5x_3 - e = 36 \\ & x_1, x_2, x_3, s, e \geq 0 \end{cases}$$

In matrix form, if

$$\vec{c}^T = \begin{bmatrix} 2 & 3 & 6 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ s \\ e \end{bmatrix}, \quad A = \begin{bmatrix} -7 & 5 & -6 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 9 & 2 & 5 & 0 & -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 9 \\ 3 \\ 36 \end{bmatrix}$$

then

$$\begin{cases} \text{Maximize } \vec{c}^T\vec{x} \\ \text{Subject to } A\vec{x} = \vec{b} \\ \vec{x} \geq 0 \end{cases}$$